

## EE 115 Homework 7

Consider a FM signal  $u_{FM}(t) = A_c \cos(2\pi f_c t + \theta(t))$  with  $\theta(t) = \theta(0) + 8\pi \int_0^t m(\tau) d\tau$ . (The time  $t$  is in second.) Apply Carson's rule to determine the bandwidth  $W$  of the FM signal for each of the following cases:

- 1) **(20 points)** The bandwidth of  $m(t)$  is 10Hz and the maximum of  $|m(t)|$  is 20.
- 2) **(20 points)**  $m(t) = 2 \cos(10\pi t)$ .
- 3) **(20 points)**  $m(t)$  has the spectrum  $M(f) = \text{rect}(f/3)$  where  $f$  is in Hz.
- 4) **(20 points)**  $m(t) = \int_0^t 2 \cos(10\pi \tau) d\tau$ .
- 5) **(20 points)**  $m(t) = \frac{d}{dt}[2 \cos(10\pi t)]$ .